

CHE 435/525 *Process Systems Analysis and Control*

Lecture 14: **Closed-Loop Transfer Functions and Closed-Loop Stability**

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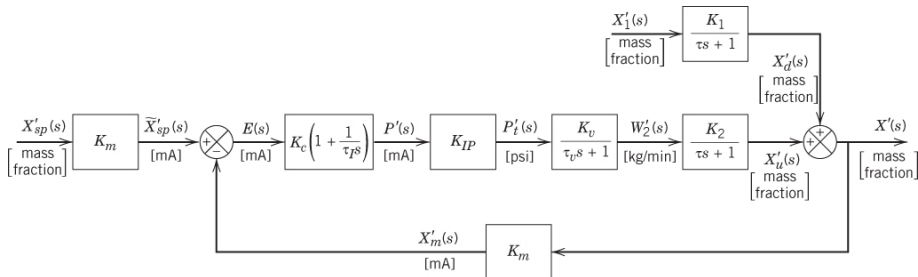


Recapture: Feedback Loop

- PID controller: A commonly used form

$$\frac{P'(s)}{E(s)} = G_c(s) = K_c \left(1 + \frac{1}{\tau_I s} + \tau_D s \right)$$

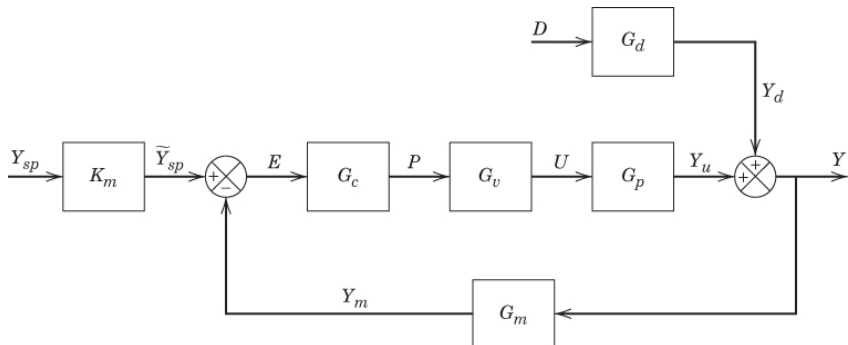
- Feedback Loop: Blending example



Learning objectives of this lecture

- ① Reduce the loop to derive **closed-loop transfer functions**
- ② Calculate **closed-loop responses** to setpoint changes and disturbances
- ③ Determine **offset** and **closed-loop stability**

Block diagram of a feedback control system



- G_p : process transfer function (plant), G_d : disturbance t.f.
- G_c : controller, G_v : valve, G_m : measurement (sensor)
- K_m : gain of G_m
- If the sensor and valve are fast enough, we may assume $G_m = 1$ (and hence $K_m = 1$) and $G_v = 1$

Simplifying the loop

- The closed-loop system is a system from $\underline{Y_{sp}(s)}$ and $\underline{D(s)}$ to $Y(s)$
- Question: What are $Y(s)/Y_{sp}(s)$ and $Y(s)/D(s)$ – **closed-loop transfer functions**?
- Answer: Calculate on the block diagram
 - ① $Y = G_p U + G_d D$
 - ② $U = G_v P = G_v G_c E = G_v G_c (\tilde{Y}_{sp} - Y_m) = G_v G_c (K_m Y_{sp} - G_m Y)$
 - ③ $Y = G_p G_v G_c (K_m Y_{sp} - G_m Y) + G_d D$
 - ④ Solve Y !

$$Y = \frac{G_p G_v G_c K_m}{1 + G_p G_v G_c G_m} Y_{sp} + \frac{G_d}{1 + G_p G_v G_c G_m} D$$

- Call $G_{OL} = G_p G_v G_c G_m$ as the **open-loop transfer function**. The denominators of the closed-loop transfer functions are both $1 + G_{OL}$.

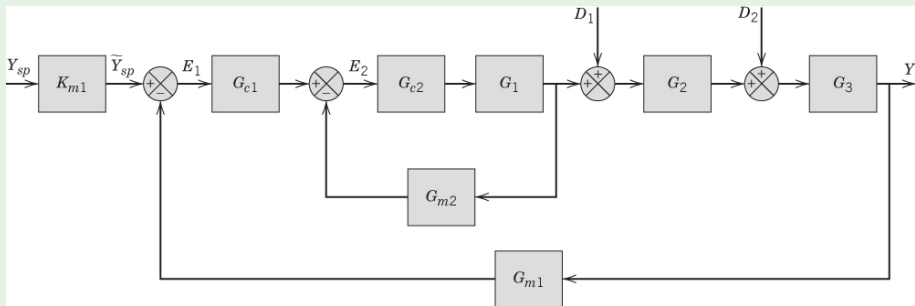
**General rules for loop simplification

- Whenever there is a feedback loop on the block diagram, with Z_i entering the loop and Z exiting the loop:

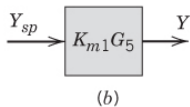
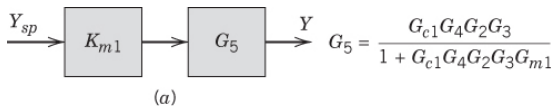
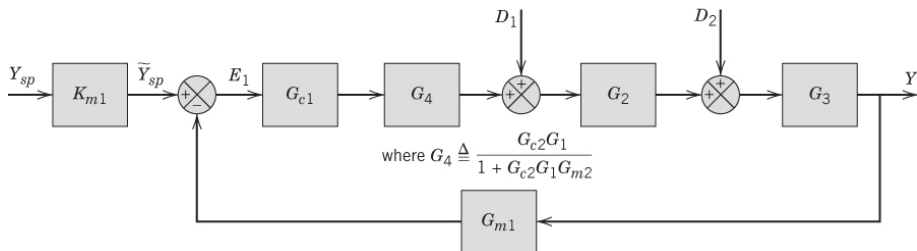
$$\frac{Z}{Z_i} = \frac{\Pi_f}{1 + \Pi_e}$$

- ▶ Π_f : product of the t.f.s on the **forward path** from Z_i to Z
- ▶ Π_e : product of **every** t.f. in the (negative) feedback loop

Complex control system [Seborg Example 11.1]



**General rules for loop simplification



Example: First-order process with P control

- Consider a first-order plant $G_p(s) = \frac{K_p}{\tau s + 1}$, $G_d(s) = \frac{K_d}{\tau s + 1}$
- Assume $G_v(s) = K_v$, $G_m(s) = K_m$ (fast valve and sensor)
- If we use a P-controller $G_c(s) = K_c$, what will be the closed-loop TFs?
- Answer:

$$\begin{aligned}\frac{Y}{Y_{sp}} &= \frac{G_p G_c G_v K_m}{1 + G_p G_c G_v G_m} = \frac{\frac{K_p}{\tau s + 1} K_c K_v K_m}{1 + \frac{K_p}{\tau s + 1} K_c K_v K_m} = \frac{\frac{K_{OL}}{\tau s + 1}}{1 + \frac{K_{OL}}{\tau s + 1}} \\ &= \frac{K_{OL}}{\tau s + 1 + K_{OL}} = \frac{K_1}{\tau_1 s + 1}\end{aligned}$$

$$\text{where } K_{OL} = K_p K_c K_v K_m, K_1 = \frac{K_{OL}}{1 + K_{OL}}, \tau_1 = \frac{\tau}{1 + K_{OL}}$$

- and also

$$\frac{Y}{D} = \frac{G_d}{1 + G_p G_c G_v G_m} = \frac{\frac{K_d}{\tau s + 1}}{1 + \frac{K_p}{\tau s + 1} K_c K_v K_m} = \dots = \frac{K_2}{\tau_1 s + 1}$$

$$\text{where } K_2 = \frac{K_d}{1 + K_{OL}}$$

Example: First-order process with P control

- Response to setpoint: $\frac{Y}{Y_{sp}} = \frac{K_1}{\tau_1 s + 1}$ where $K_1 = \frac{K_{OL}}{1 + K_{OL}}$, $\tau_1 = \frac{\tau}{1 + K_{OL}}$
 - ▶ For a step change of magnitude M in the setpoint, the response is

$$y(t) = K_1 M (1 - e^{-t/\tau_1})$$

- ▶ **Offset** $= M - K_1 M = \frac{1}{1 + K_{OL}} M$
 - ▶ If $K_{OL} = K_p K_c K_v K_m \gg 1$, then the offset is small but **never reaches 0**
- Response to disturbance: $\frac{Y}{D} = \frac{K_2}{\tau_1 s + 1}$ where $K_2 = \frac{K_d}{1 + K_{OL}}$
 - ▶ For a step change of magnitude M in the disturbance, the response is

$$y(t) = K_2 M (1 - e^{-t/\tau_1})$$

- ▶ If $K_{OL} \gg 1$, the **gain from disturbance** is small but **never reaches 0**
- Observation: **P-controller is not offset-free!**

Example: First-order process with PI control

- Still consider $G_p(s) = \frac{K_p}{\tau s + 1}$, $G_d(s) = \frac{K_d}{\tau s + 1}$
- Assume $G_v(s) = K_v$, $G_m(s) = K_m$ (fast valve and sensor)
- If we use a PI-controller $G_c(s) = K_c(1 + \frac{1}{\tau_I s})$, what will be the closed-loop TFs?
- Answer:

$$\begin{aligned}\frac{Y}{Y_{sp}} &= \frac{G_p G_c G_v K_m}{1 + G_p G_c G_v G_m} = \frac{\frac{K_p}{\tau s + 1} K_c (1 + \frac{1}{\tau_I s}) K_v K_m}{1 + \frac{K_p}{\tau s + 1} K_c (1 + \frac{1}{\tau_I s}) K_v K_m} \\&= \frac{K_p K_c K_v K_m (\tau_I s + 1)}{(\tau s + 1) \tau_I s + K_p K_c (\tau_I s + 1) K_v K_m} = \frac{K_{OL} (\tau_I s + 1)}{\tau \tau_I s^2 + (1 + K_{OL}) \tau_I s + K_{OL}} \\&= \frac{\tau_I s + 1}{\frac{\tau \tau_I}{K_{OL}} s^2 + \frac{1 + K_{OL}}{K_{OL}} \tau_I s + 1} \quad (\text{gain} = 1 \text{ by evaluating at } s = 0)\end{aligned}$$

- Observation: **PI-controller is offset-free!**

Closed-loop stability and characteristic equation

- Now we can calculate closed-loop responses based on a given controller ($G_c \rightarrow Y/Y_{sp}, Y/D$)
- The actual question is the converse – **how to design a controller to achieve desirable response?**
- Most basic requirement for control: **closed-loop stability**

$$\frac{Y}{Y_{sp}} = \frac{G_p G_v G_c K_m}{1 + G_{OL}} \text{ and } \frac{Y}{D} = \frac{G_d}{1 + G_{OL}} \text{ must be stable TFs!}$$

General stability criterion

The feedback control system is stable if and only if all the roots of the **characteristic equation**

$$1 + G_{OL}(s) = 0 \quad (1)$$

have negative real parts.

Example: P-controller for an unstable first-order system

- **Question:** Given $G_p(s) = \frac{K_p}{\tau_p s - 1}$ and $G_v = G_m = 1$, if a P-controller $G_c(s) = K_c$ is to be used, what is the range for K_c to guarantee closed-loop stability?

- **Answer:** The characteristic equation is

$$1 + G_{OL}(s) = 1 + \frac{K_c K_p}{\tau_p s - 1} = 0$$
$$\tau_p s - 1 + K_c K_p = 0$$

- In order that the root has negative real part, we need $K_c K_p > 1$.
- **Insight:** Feedback controllers can stabilize unstable processes, if using sufficiently large controller gains.

Example: P-controller for a non-minimum-phase system

- **Question:** Given $G_p(s) = \frac{K_p(1-\tau_0s)}{(1+\tau_1s)(1+\tau_2s)}$ and $G_v = G_m = 1$, if a P-controller $G_c(s) = K_c$ is to be used, what is the range for K_c to guarantee closed-loop stability?

- **Answer:** The characteristic equation is

$$1 + G_{OL}(s) = 1 + \frac{K_c K_p (1 - \tau_0 s)}{(1 + \tau_1 s)(1 + \tau_2 s)} = 0$$

$$(1 + \tau_1 s)(1 + \tau_2 s) + K_c K_p (1 - \tau_0 s) = 0$$

$$\tau_1 \tau_2 s^2 + (\tau_1 + \tau_2 - K_c K_p \tau_0) s + K_c K_p = 0$$

- Note that the two roots (say s_1, s_2) satisfies $s_1 + s_2 = -\frac{\tau_1 + \tau_2 - K_c K_p \tau_0}{\tau_1 \tau_2}$ and $s_1 s_2 = \frac{K_c K_p}{\tau_1 \tau_2}$
- In order that the root has negative real part, we need

$$-\frac{\tau_1 + \tau_2 - K_c K_p \tau_0}{\tau_1 \tau_2} < 0 \text{ and } \frac{K_c K_p}{\tau_1 \tau_2} > 0 \Rightarrow 0 < K_c K_p < \frac{\tau_1 + \tau_2}{\tau_0}$$

- **Insight:** Non-minimum-phase requires a “not-too-large” controller gain.

Far from enough...

- Analysis for **higher-order processes** and **controllers with multiple parameters** (e.g., PID)
- Analysis in the presence of time delays
- Approaches for designing **generic controllers** beyond “PID”